Same 2,000 validation samples | 256-D fusion space, no PCA/t-SNE
L2-normalize samples, average by true class, compare mean directions

**Binary (original binary head)**

| | Angry | Bored | Calm | Fear | Happy | Love | Sad |
|---|---|---|---|---|---|---|---|
| Angry | 1.000 | 0.993 | 0.793 | 0.996 | 0.658 | 0.691 | 0.991 |
| Bored | 0.993 | 1.000 | 0.845 | 0.994 | 0.726 | 0.754 | 0.996 |
| Calm | 0.793 | 0.845 | 1.000 | 0.828 | 0.961 | 0.972 | 0.853 |
| Fear | 0.996 | 0.994 | 0.828 | 1.000 | 0.703 | 0.733 | 0.995 |
| Happy | 0.658 | 0.726 | 0.961 | 0.703 | 1.000 | 0.998 | 0.726 |
| Love | 0.691 | 0.754 | 0.972 | 0.733 | 0.998 | 1.000 | 0.758 |
| Sad | 0.991 | 0.996 | 0.853 | 0.995 | 0.726 | 0.758 | 1.000 |

**V4**

| | Angry | Bored | Calm | Fear | Happy | Love | Sad |
|---|---|---|---|---|---|---|---|
| Angry | 1.000 | 0.749 | 0.605 | 0.783 | 0.521 | 0.547 | 0.851 |
| Bored | 0.749 | 1.000 | 0.684 | 0.624 | 0.633 | 0.591 | 0.850 |
| Calm | 0.605 | 0.684 | 1.000 | 0.679 | 0.736 | 0.735 | 0.697 |
| Fear | 0.783 | 0.624 | 0.679 | 1.000 | 0.544 | 0.601 | 0.805 |
| Happy | 0.521 | 0.633 | 0.736 | 0.544 | 1.000 | 0.960 | 0.631 |
| Love | 0.547 | 0.591 | 0.735 | 0.601 | 0.960 | 1.000 | 0.643 |
| Sad | 0.851 | 0.850 | 0.697 | 0.805 | 0.631 | 0.643 | 1.000 |

**V5**

| | Angry | Bored | Calm | Fear | Happy | Love | Sad |
|---|---|---|---|---|---|---|---|
| Angry | 1.000 | 0.854 | 0.706 | 0.818 | 0.644 | 0.646 | 0.902 |
| Bored | 0.854 | 1.000 | 0.759 | 0.726 | 0.736 | 0.705 | 0.920 |
| Calm | 0.706 | 0.759 | 1.000 | 0.755 | 0.825 | 0.825 | 0.790 |
| Fear | 0.818 | 0.726 | 0.755 | 1.000 | 0.645 | 0.672 | 0.828 |
| Happy | 0.644 | 0.736 | 0.825 | 0.645 | 1.000 | 0.975 | 0.731 |
| Love | 0.646 | 0.705 | 0.825 | 0.672 | 0.975 | 1.000 | 0.721 |
| Sad | 0.902 | 0.920 | 0.790 | 0.828 | 0.731 | 0.721 | 1.000 |

**V6**

| | Angry | Bored | Calm | Fear | Happy | Love | Sad |
|---|---|---|---|---|---|---|---|
| Angry | 1.000 | 0.854 | 0.605 | 0.840 | 0.539 | 0.551 | 0.900 |
| Bored | 0.854 | 1.000 | 0.673 | 0.729 | 0.638 | 0.614 | 0.912 |
| Calm | 0.605 | 0.673 | 1.000 | 0.670 | 0.806 | 0.804 | 0.705 |
| Fear | 0.840 | 0.729 | 0.670 | 1.000 | 0.555 | 0.594 | 0.855 |
| Happy | 0.539 | 0.638 | 0.806 | 0.555 | 1.000 | 0.977 | 0.625 |
| Love | 0.551 | 0.614 | 0.804 | 0.594 | 0.977 | 1.000 | 0.630 |
| Sad | 0.900 | 0.912 | 0.705 | 0.855 | 0.625 | 0.630 | 1.000 |

Cosine similarity of class centers (fixed 0–1 scale)
1.0
0.8
0.6
0.4
0.2
0.0